

According to the International Energy Agency (IEA), China has also established a near monopoly on battery components production, supplying almost 85% of cathode active materials, including NMC (Nickel, Manganese, Cobalt) and LFP chemistries, and over 90% of anode active material production, predominantly graphite.

Cell manufacturers have to keep costs low for automakers and ESS (energy storage system) players. India-made cells are expected to be 25-30% costlier than China. "If cell makers buy everything from China at the cheapest price that they get and not buy anything locally, they are still 25% more expensive than a Chinese cell," says Handa.

Without Indian customers, Epsilon is targeting 80% sales from exports to the United States. In the short term, the US is the real market because of energy storage and data centres. Korean giants like LG and Samsung have also set up cell manufacturing capacity in the US.

Epsilon currently makes graphite anode in India at its 2,000-tonne plant, which can cater to 2 GWh of cell capacity. It is looking to scale it up to 30,000 tonnes by mid-2028. "Even by that time, India will not be producing 30 GW of cells, so we will export a lot of it to customers in the US," says Handa.

The company has also started a pilot line for cathode in Germany. "We are looking to build a 30,000-tonne plant in Karnataka. That will break ground before the end of the year and commission by early 2028. The 30,000 tonnes of cathode that we plan to make will cater to 15 GWh of cell making. For that, we will require 5,000 tonnes of lithium carbonate," says Handa,

adding that Epsilon will source lithium from China, Australia and South America.

RECYCLING TO THE RESCUE

Battery recycling can reduce India's dependence on critical minerals in a meaningful manner. In base metals such as aluminum, copper and iron, anywhere between 30% and 50% of the production comes from recycled materials, says Lohum CEO Rajat Verma. Over the long term, anywhere

bility, ESS, consumer electronics and production scrap by 2030.

The battery recycling market will be \$10 billion by 2030, says Attero's Gupta. "Even today, globally, 80% of batteries coming back for recycling are from production waste, where these batteries are manufactured. Every gigafactory globally has 10% to 25% of capacity as production waste," he says.

Under the government's ₹7,280 crore Rare Earth Permanent

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between 30% and 50% production of critical minerals will come via recycling in the country, he says.

It's already being seen in the lead-acid battery industry where most of the lead that is required for manufacturing comes from recycling, says Amara Raja's Gourineni. "Recycling is dominating that value chain."

Recycling can eventually provide a meaningful supply of critical minerals like lithium, nickel, cobalt and rare earths. "For a metal like cobalt, we can meet 60% demand from recycling. For metals like lithium carbonate, it is 70%, and for rare earths, 75%," says Nitin Gupta, co-founder and CEO of battery recycler Attero.

According to NITI Aayog, India will have 92,000 tonnes of end-of-life lithium-ion battery from mo-

Magnet scheme, both Attero and Lohum have plans to get into magnet production as demand for rare earth magnets is projected to double from 4,000 tonnes in 2025 to 8,000 tonnes by 2030. Unlike lithium, where India is entirely dependent on imports, the country has reserves of rare earths. State-owned IREL (India) Ltd, the sole domestic producer of rare earth magnets, runs an extraction plant in Odisha and a refining unit at Aluva in Kerala. Another state-run firm KABIL (Khanij Bidesh India Ltd) has secured five lithium blocks in Argentina.

"India can break China's hegemony globally by becoming a big battery recycling market," adds Gupta. **BT**

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